Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2410)

Key

SECTION – I

1. B
2. D
3. D
4. C
5. D
6. D
7. B
8. B
9. C
10. C
11. D
12. B
13. D
14. B
15. B
16. A
17. B
18. D
19. 4123
20. A
21. D
22. 3241
23. 4231
24. C

SECTION – II

1. 12
2. A
3. B
4. 3
5. C
6. D
7. C
8. 240
9. 220
10. A
11. C
12. A
13. A
14. B
15. B
16. A
17. C
18. C
19. 7
20. 1

SECTION – III

1. 530
2. 43.2
3. B
4. D
5. 12
6. 2592
7. C
8. B
9. 4220
10. A
11. 146
12. B
13. D
14. A
15. C
16. D
17. C
18. B
19. 34
20. 21
21. D
22. A

Solutions

SECTION – I

Solutions for questions 1 to 3:

Number of words and Explanatory notes for RC:

Number of words: 387

1. Option A: This can be inferred from 'Meaning-seeking creatures need stories to deal with the unknown and provide guidance as to where we came from.' Our stories help us deal with the unknown. Option A is not the answer.
Option B: This is incorrect. Morality is enforced by rituals. This has been mentioned in the second para. Also, rituals codify belief in a way that enforces morality and vouches for shared social activities. These are two separate ideas. Morality doesn't enable our commitment to social activities. Rituals enforce morality and vouch for shared social beliefs. Option B is the answer.
Option C: Consider this: '***Meaning-seeking creatures need stories to deal with the unknown and provide guidance as to where we came from and where we're going***.' We look to mythology to answer these questions. So, it can be inferred that we look towards mythology for a sense of purpose and identity. Option C is not the answer.
Option D: Myths are a solution for that which is beyond practical, scientific and logical thought according to the last line of the second para. So, it can be inferred that science and logic cannot explain or explain away/justify everything in life. Option D is not the answer. Choice (B)

2. Option A: This is negative about psychology, but the first sentence is not entirely negative. It points out that psychology has adopted a new function, pushing the idea of personal enlightenment. It emphases society's pressure, but it is not that society uses psychology to pressure us. Hence, Option A is not the answer.
Option B: The author discusses how psychology takes up the disenchanted space. However, the author doesn't separate enchanted and disenchanted individuals into two groups. Hence, Option B is not the answer.

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Option C: The popularity of psychology has been discussed later. What is apparent in the self-help literature is not related to whether psychology is popular or not. It is about how contemporary psychology has changed its approach. Option C is not the answer.

Option D: This is what the author was referring to: 'the rhetoric of personal enlightenment'. It is obvious from self-help literature that psychology ***now*** offers people a chance at salvation, by pushing the idea of personal enlightenment through its therapeutic practices. Option D is the answer. Choice (D)

3. Option A: While reflective belief is satisfied by explanations, non-reflective belief is not specifically satisfied by 'physical action.' It is more about belief and commitment to rituals. Option A is not the answer.

   Option B: While non-reflective belief is affective and hence emotional, the former is cognitive, and related to thoughts, logic and principles, not just introspection. Hence, Option B is close but not the answer.

   Option C: While rituals involve actual steps of practice, explanations for our reflective belief aren't just about 'theory'. Theory as a term is vague and cannot apply here. Option C is not the answer.

   Option D: The mythology of psychology provides explanations through causal-predictive principles and origin stories for our reflective belief. So, the former is about receiving answers. For non-reflective belief, myth is enacted in a set of rituals which are about making commitments in the act of belief. This is further understood from '***Together, mythology and its practices in ritual provide the platform for creating meaning through making commitments in the act of belief. It does so by expressing, enhancing and codifying belief.***' So, the latter is about ritual, and by extension, following a codified process. Hence, Option D is apt and is the answer.

   Choice (D)

Solutions for questions 4 to 8:

Number of words and Explanatory notes for RC:

Number of words: 524

4. The author provides the definition of axiom with respect to language, thought and culture as "an assumption that informs all research on language in social contexts". Here, the assumption is related to the nature of the relationship between the three aspects under discussion – language, culture and thought.

   Option A: They are three different entities that influence each other. They are not interchangeable, and they obviously cannot take one another's positions especially considering that thought and culture are non-linguistic entities. Option A is not the answer.

   Option B: Even though it is true, according to the last para of the passage, this is not the assumption. The assumption or the axiom is that thought, culture and language are overlapping entities, and it is not possible to precisely differentiate the influences of each on the other. Option B is not the answer.

   Option C: This is the axiom that is used as an assumption as understood from 'it is also possible that the relative strength of the influence of these three overlapping entities could never be specified with any precision.' We cannot clearly define the boundaries of the relationship between the three entities – language, thought and culture, and how they influence each other. Option C is the answer.

   Option D: While they form a multidirectional relationship, the assumption is that it is not empirically testable. Hence, the relationship cannot be clearly defined and the relationship is more like an axiom. Hence, Option D is not the answer. Choice (C)

5. The 'strong version' of the hypothesis is that 'the particular language you speak rigidly structures your thought in an inescapable manner.' After proving the option false, it should be compatible to this view. So, the option as it stands has to contradict this view.

   Option A: This option doesn't contradict the strong version. It directly supports it by saying language has a strong influence on our thoughts. Hence, Option A is not the answer.

   Option B: This option doesn't contradict the strong version. It supports it by asserting that thoughts are rigidly constructed based on the language we speak. Option B is not the answer.

   Option C: If contradicted, this leads to the argument that human thought does need language to be expressed. However, that doesn't say whether thoughts are shaped by language or not. Hence, Option C is irrelevant to the given argument. Option C is not the answer.

   Option D: If proven to be false, it leads to the argument that no fundamental factors other than language influence our thoughts. So, that is compatible with the strong version. Hence, Option D is the answer. Choice (D)

6. The author provides two reasons for calling the Sapir-Whorf hypothesis a misnomer.

   (i) Sapir and Whorf never co-authored anything, even though Whorf was influenced by Sapir.

   (ii) they never framed their research using these terms.

   Option A: Firstly, there wasn't a fixed hypothesis framed by respected scholars. The scholars never framed their research either using terms from the realm of science. So, this option is inaccurate and also doesn't explain why the term 'misnomer' was used. Option A is not the answer.

   Option B: Some highly respected scholars formulated empirically testable hypothesis based on the insights of Boas, Sapir and Whorf. However, the option doesn't explain the use of the word 'misnomer'. Option B is not the answer.

Option C: While this option is partially true, it doesn't explain why the hypothesis was referred to as a misnomer. Option C is not the answer.
Option D: This explains why the author called it a misnomer. Sapir and Whorf never worked together to frame a hypothesis. So, calling a hypothesis by their name is a wrong representation. Hence, it is a misnomer. Option D is the answer. Choice (D)

7. Consider this: '***Everything is linguistically mediated, they contend, so it is impossible to separate something called "language" from the two allegedly non-linguistic entities called "thought" and "culture."'*** To weaken this argument, we have to show that it is possible to separate language from the other two entities. We can show that thought and culture can be mediated without language.
Option A: Thought and culture take shape only through language. If this is true, it means thought and culture are indeed linguistically mediated. This supports the axiom view, it doesn't weaken it. Option A is not the answer.
Option B: If true, this goes against the axiom view that thought and culture can only ever be linguistically mediated. Here, it says that art/paintings can also be a channel for thought and culture, which can, therefore, be non-linguistically mediated. Hence, this option weakens the argument. Option B is the answer.
Option C: This strengthens the axiom view by explaining that it is indeed true, that thought and culture can only be mediated through language. Hence, Option C is not the answer.
Option D: Even if it is true, it doesn't really show that thought and culture are independent of language and need not be only linguistically mediated. Hence, Option D is not the answer.
Choice (B)

8. Option A: The author's biggest concern/arguments in the passage are not about tangible influence of language on thought and culture as understood from ***'The relative strengths of the relationships remain unspecified and perhaps unknowable.'*** In fact, this sentence shows that the author doesn't believe a tangible analysis is possible as the author says it is perhaps unknowable to know what the exact influence of each on the other (language, thought and culture) is. So, the influence of one language versus that of another language might not be possible to differentiate. Hence, Option A is
Option C: Such a study is not quite effective because the author already clearly establishes that both affect each other, and then there is a third parameter in the form of culture. Option C is not the answer.
Option D: While all three factors influence each other, the discussion never focuses on what influences the other more. Also, such a study would practically not be possible in a tangible way. We can only understand whether the influence varies, not by how much. Option D is not the answer. Choice (B)

Solutions for questions 9 to 11:

Number of words and Explanatory notes for RC:

Number of words: 409

9. Option A: This can be understood to be true from "In the layers of culture theory, Trompenaars distinguished three levels, ranging from the most to the least explicit." Option A is not the answer.
Option B: In iceberg theory, there are external and internal elements, and the internal elements include both rituals and customs, which are labelled formal and unconsciously acquired elements, which are labelled informal. Option B is not the answer.
Option C: This is not accurate. According to the onion skin model, values are part of the less visible layer as understood from 'the least visible part, the heart of a culture, consists of its values.' This option mentions that values as more visible. Option C is the answer.
Option D: The second level of layers of culture model is formed by norms of social behaviour, while the third level includes unconsciously acquired forms of behaviour. So, this grouping is correct. Option D is not the answer. Choice (C)

10. Option A: This is possible as understood from the fact that every model talks about the elements of culture that are inaccessible from outside. It means, they can be studied or inspected closely. Otherwise, their existence wouldn't have been observable in the first place. Option A is not the answer.
Option B: The individualism and the typologies of behaviour are influenced by the experience of a group or the crisis a group has survived or solved. This has already been explained in the passage. It also shows that individuals subconsciously imbibe some elements and these elements of a culture

not the answer.

Option B: The author says that ***'This sort of multidirectional relationship is best conceived of as an axiom – an assumption that informs all research on language in social contexts.'*** Since, the multidimensional relationship between language, culture and thought informs/guides research about how language functions in social contexts, this will be a study the author will approve of as understood from the passage. This multidimensional relationship can influence such research and is well-worth studying further. Option B is the answer.

are not explicitly visible from outside. Option B is not the answer.

Option C: This has not been discussed in the passage. However, since the passage is about business management, and since we are trying to understand culture for practical applications, this is a relevant question that needs to be answered. Option C is the answer.

Option D: This is possible. We already know that from 'which are reflected in the distinct dimensions of culture that can be observed today...' (last sentence of the passage). Option D is not the answer. Choice (C)

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11. Option A: This can be understood from 'the most profound base of a culture is made up of a series of unconscious elements, which form what at one definite time was the group response to a concrete problem that implied the survival or otherwise of the group.' The unconscious elements were the group response to a concrete problem. Option A is not the answer.

Option B: That cultures are layered according to all the models can be understood from 'the three models have in common is that they define culture in levels, some of which are external and some of which are internal.' There are internal and external levels in a culture according to all the models. Option B is not the answer.

Option C: Behaviours of a community are influenced subconsciously by intrinsic elements of its culture because the author says that 'These responses conditioned typologies of behavior in relation to time or other people.' These responses are the responses of a group at the time of a crisis, for survival or to solve a concrete problem. Option C is not the answer.

Option D: The author says that 'they define culture in levels, some of which are external and some of which are internal, some visible from outside and others that cannot be recognized from outside.' So, some cultures cannot be understood from the outside. The author doesn't say that the culture cannot be fully understood, no matter what. Option D cannot be inferred. Option D is the answer. Choice (D)

Solutions for questions 12 to 16:

the meaning of theory is normative – theories of what ought to be.' Option D is not the answer. Choice (B)

13. Option A: This can be inferred from 'theory means a rigorous testing of predictive theorems or hypotheses using observable and comparable data.' Since the same is not possible for public administration theories, because of the problems involved in recognizing patterns, this option is indeed a difference. Option A is not the answer.

Option B: This can be understood from 'Theory in the natural or physical sciences can claim considerable accuracy in representing reality.' So, there is better quality of data in case of physical sciences. Option B is not the answer.

Option C: Science has easily comparable natural phenomena where measurements are advanced, according to the author. However, there are a lot more problems with public administration theories, in the social world. Hence, this is a valid difference. Option C is not the answer.

Option D: While the author says that science has predictive and comparable theorems that can be tested, there is no comparison of 'the number of theorems' that exist in each field. Hence, Option D is the answer. Choice (D)

14. Option A: The author doesn't suggest that every theorist has a different rigor to the process. The author also doesn't suggest that factual evidence changes based on the theorist. The subjectivity is because of the way a theory is constructed by a theorist based on concepts, definitions and metaphors (factual material). Option A is not the

Number of words and Explanatory notes for RC:

Number of words: 519

12. The passage talks about three meanings of theory, of which the first one cannot be applied to theory of public administration, while the other two are in line with the theory of public administration. We need to identify the option which does not fall under any of the three meanings discussed in the passage.
Option A: This can be understood from 'First, in the natural and physical sciences, theory means a rigorous testing of predictive theorems or hypotheses using observable and comparable data.' Option A is not the answer.
Option B: This option talks about how we can identify a theory to be useful. It is not mentioned as one of the three fundamental definitions/meanings of a theory. Option B is the answer.
Option C: This is the second meaning of theory based on how it is perceived in the world of social sciences, where data is harder to collate, and theorists collect factual evidence and present a subjective evaluation of reality. Option C is not the answer.
Option D: This is the third meaning of theory as understood from 'Third, in public administration answer.
Option B: This explains why the author calls the understanding 'subjective' as understood from 'To be sure, this understanding is subjective, ***because it was constructed by the theorist.'*** The underlying assumption is that the theorist brings subjectivity because every theorist has a different way of constructing the theory. Option B is the answer.
Option C: The author doesn't complain that the theories do not go beyond intuition. The author says that the theory is based on rigorous and intuitive observation of behaviour and systems. Also, this doesn't form the basis for calling the understanding subjective. Option C is not the answer.
Option D: Public administration decisions and actions are based on fundamental assumptions which in turn are described by the theorist. So, this option is factually incorrect and also is not the basis for why the author calls the understanding subjective. Option D is not the answer.
Choice (B)

15. Option A: While this may be true, it doesn't explain why the author calls these theories, theories of politics. The author does so because these theories guide political processes, specifically allocation of public goods. Option A is not the answer.

Option B: As the author says, '***Theories of public administration guide the authoritative allocation of public goods.'*** This explains why the public administration theories are also political, because they have an influence over the dispensing of political duties by the state. Option B is the answer.
Option C: The public administration theorist doesn't just observe but also guide the authoritative allocation. According to the author, the theories are political as well because they are the basis for the authoritative allocation. Hence, Option C is not the answer.
Option D: Theories of public administration guide the authoritative allocation of public goods. Hence, it is political as well. It is not political because the theorist has to evaluate how the state does it. The theory guides the state on how to do it. Option D is not the answer. Choice (B)

that language doesn't accurately depict the brain states that generated it in the first place (inner brain states) and hence we are fooled at times. Option B is the answer.
Option C: Language doesn't actually always fall short of expressing our actual brain state. Language can fool us because sometimes it doesn't fully correspond to the brain states that generated it. It should be noted here that 'falling short' is different from 'inaccurately represent'. Language inaccurately represents our brain states, but it doesn't fail completely in expressing our brain states (thoughts and feelings). Option C is not the answer.
Option D: Language doesn't represent our brain states perfectly. It could mean we are aware of our brain state but don't have the right words to articulate it. It could also mean we don't know what our brain state is correctly. Hence, this cannot be decided for sure. Option D is not the answer.

16. Consider this: '***The test of a theory's usefulness is often its criteria in selecting and classifying facts, and if these are accurate, the theory will enhance understanding, guide research***, and powerfully describe, explain, and predict.'
Option A: The author says that a theory will guide understanding if it is useful and a test for its usefulness is the accuracy of the criteria it uses to select and classify facts, and pick the most important ones. Hence, Option A is the answer.
Option B: A useful theory does all this along with enhancing understanding and guiding research. A theory doing these is not a necessary criterion for a theory to be useful (which is a necessary condition for a theory to enhance understanding and guide research). Option B is not the answer.
Option C: A theory is useful if it organises and classifies data based on specific criteria, not based on its usefulness. Option C is a misrepresentation of the author's opinion about what a useful theory must do. Option C is not the answer.
Option D: The author doesn't say that a theory's usefulness has to be proven factually. The author says that it can be understood based on the criteria used by such a theory to sift through facts. Hence, Option D although partially correct is, to some extent, misleading. Option D is not the answer. Choice (A)

Solutions for questions 17 and 18:

17. Option A: Language and conscious brain states are closely associated according to the para. However, we don't know whether sub-conscious brain states can generate language from this information. That language can provide a perfect read-out of the underlying brain states could also mean language cannot really articulate our conscious brain states. Option A is not the answer.
Option B: This is definitely true as understood from 'the close relationship between it and conscious brain states ***does not mean that language can provide a perfect read-out of the underlying brain states*** that generate it.' The bold part shows

Choice (B)

18. Option A: If this is true, it weakens the argument of the Bitcoin advocates, as it shows that people are valuing gold despite not knowing who else does. Option A is not the answer.
Option B: While there is a corelation signified here, this option neither strengthens nor weakens the argument of Bitcoin advocates as it doesn't tell us what causes people to value gold. It merely tells us that gold is valued similarly by those in the same economic situation. Option B is not the answer.
Option C: This explains why gold is valued and it offers a tangible reason that is not based on other people. This option weakens the argument of the Bitcoin advocates. Option C is not the answer.
Option D: If true, it shows that people's valuation of gold is based on their perception of other people's valuation of gold, in the form of higher social status. Hence, Option D is the answer. Choice (D)

Solution for question 19:

19. Sentence 1 points to 'this technique', something that helps them capitalize on the low-pressure zones created by cyclists in front of them.
Sentence 2 explains how 'this technique' allows cyclists to capitalise. It elaborates on the point made in 1 which merely touches upon the advantage. Hence, 12 is a block.
Sentence 3 is a factual statement that proves the extent of advantage provided by drafting and puts a number next to it. The number makes sense only if we already understand what advantage the author is talking about. That has been explained in the 12 block. So, 3 follows 12.
Sentence 4 introduces the concept of drafting and defines it. It provides the basis for 'this technique' in 12. So, 4 comes ahead of the 123 block. Hence, 4123 is the order.

Ans: (4123)

Solution for question 20:

writers and producers is the sentence talking about.

20. Option A: This is the right place for the sentence. In this sentence, the tone is one of a sense of confusion on the part of the Oppenheimer team as understood from 'scratching their heads.' The first sentence of the given para continues the tone explaining why it is difficult for Oppenheimer to compete with Barbie. Option A is the answer.

Option B: This blank is preceded by 'particularly the film' which means that the film has already been introduced by then. So, the bold statement won't fit in this blank. It has to precede the sentence that precedes blank 2. Option B is not the answer.

Option C: This blank cannot be populated considering the sentence that precedes it and the sentence that follows it have the same tone. Also, the sentence after this blank ends with 'make the comparisons all the more stark.' So, the preceding sentence must also have comparisons. Option C is not the answer.

Option D: Oppenheimer's team scratching their heads won't make sense here anymore, because by here the solution has already been found by the internet, and explained. Option D is not the answer. Choice (A)

Solution for question 21:

21. Option A: It has been mentioned that China will try to dominate Northeast Asia militarily. Nowhere has it been mentioned that Chine will dominate the United States militarily. Hence, Option A is not the answer.

Option B: In order to form a balancing coalition with China's neighbors to contain China's growing power, it is not necessary that the US currently share friendly relationships with them. It could be necessity-based as well. Option B is not the answer.

Option C: This is already mentioned in the para, and the author's conclusion that the US and China will forever be adversaries doesn't directly arise from this point. Option C is not the answer.

Option D: Since hegemony only helps a country with its own security (as understood from the para), the US needs to form a balancing coalition to stop Chinese hegemony, if the US feels threatened by Chinese hegemony. This option is the underlying basis of the author's conclusion that China and the US will remain adversaries. If the US doesn't feel threatened by Chinese hegemony, there is no reason for the United States to form a coalition and try and contain China's growing power. Option D is the answer. Choice (D)

Solutions for questions 22 and 23:

22. Sentence 1 starts with a positive connector 'And'. This sentence is contextually dependent on another sentence as we need to understand which Sentence 2 is an independent sentence that discusses what Gregory House and his team do.

Sentence 3 precedes Sentence 2. This can be understood from the fact that 3 introduces the character name 'Gregory House' as well as the show name, while 2 explains it further. So, 32 is a block.

Sentence 4 points to 'this blend'. We know that 24 is a block, because 2 has the contrast 4 points to – the oddity of the medical condition and the mystery that such a story entails.

We can understand that 1 follows 4 because 1 starts with 'And' and provides an add-on to the fact that House was a well-watched show. 1 adds to it by pointing out that even the writers and producers are having fun. So, 3241 is the order.

Ans: (3241)

23. Sentence 1 starts with the connector 'still' and points to an indifference in a preceding sentence. It also points out that this indifference doesn't matter.

Sentence 2 starts with a connector 'Add to this' explaining how whatever is being discussed is confusing to the English ear.

Sentence 3 starts with the connector 'Typical' and explains the causation for something previously mentioned.

Sentence 4 is an independent sentence that introduces how difficult classical tradition in music is.

42 is a block because of the connector 'Add to' that is followed by something in the same tone as in 4 – parameters that make Indian classic music difficult to comprehend. 3 should follow 2 as 2 introduces the idea of the English ear and 3 explains the causation for why the ordinary English ear is baffled by Indian classical music. 'This indifference' in 1 fits perfectly with the 'colonial indifference' in 3. So, 31 is a block. Hence, the order is 4231.

Ans: (4231)

Solution for question 24:

24. Option A: It cannot be the first blank, because the statement preceding the blanks only talks about the protection of animals being solidified as a legal concern and the statement following it talks about 'domestic law'. So, the 'moreover' in the bold statement as a positive connector, won't make sense preceding the sentence where domestic law has been introduced. Option A is not the answer.

Option B: It cannot be the second blank because the bold statement talks about global animal law, but the sentences following this blank are examples of individual laws – examples of domestic law. Hence, Option B is not the answer.

Option C: This blank is apt, as it justifies the use of ‘moreover’. The sentence talks about ‘global animal law’ as a continuation of ‘domestic laws.’ It is followed by the conclusion. Hence, Option C is the answer.

Option D: This blank starts with ‘overall’ and has the tone of a conclusion. So, adding another sentence about global animal law after this statement doesn’t make sense. Option D is not the answer. Choice (C)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	20
Medium	4, 6, 7, 9, 17, 19, 21, 22, 23
Difficult	1, 2, 3, 5, 8, 10, 11, 12, 13, 14, 15, 16, 18, 24
Very Difficult	–

SECTION – II

Solutions for questions 1 to 4:

Given that the leaves that are provided in the table do not include Saturdays and Sundays. We need to find the dates corresponding to the Saturdays and Sundays to calculate the number of leaves for each employee.
For November, between 1st and 7th, there should be at least one Saturday and one Sunday.
Since 1st and 5th are in the table, 1st and 5th cannot be Saturday or Sunday.
Since 11th is in the table, 11th and hence, 11 – 7 = 4th cannot be a Saturday or a Sunday.
Since 1st, 4th and 5th cannot be Saturday or Sunday, the Saturday and Sunday from 1st to 7th must be either 2nd and 3rd OR 6th and 7th.
If November 6th is a Saturday, then November 27th is also a Saturday. Hence, December 4th will also be a Saturday and December 5th will be a Sunday. But December 5th cannot be a Sunday because the date is present in the table. Hence, November 6th and 7th cannot be Saturday and Sunday. November 2nd must be Saturday and November 3rd must be Sunday.

The Saturdays and Sundays in November and December are:

November		December	
Saturday	**Sunday**	**Saturday**	**Sunday**
2nd	3rd		1st
9th	10th	7th	8th
16th	17th	14th	15th
23rd	24th	21st	22nd
30th		28th	29th

From this and the leaves of the employees, we can calculate the total number of leaves and the salary paid to each employee:

Employee	Type	Monthly Salary	November		December	
			Leaves	Salary Paid	Leaves	Salary Paid
Aniket	Analyst	25000	7	19167	3	22581
Banik	Manager	70000	2	65333	4	60968
Chunky	Analyst	25000	5	20833	2	23387
Dory	Executive	100000	2	93333	2	93548
Ekant	Specialist	85000	8	62333	3	76774
Fakir	Specialist	85000	3	76500	4	74032
Gorak	Executive	100000	5	83333	7	77419

1. The maximum total *effective number of leaves* for any employee across the two months combined is 12 (for Gorak). Ans: (12)

2. The total salary (in ₹) paid to all the employees for the month of November = 420832. Choice (A)

3. The sum of the *effective number of leaves* for the seven employees in December is 25. Hence, the answer is option B. Choice (B)

4. Any employee who had at least 6 leaves in November and at least 7 leaves in December would have been paid at most 80% of their monthly salary in those months. Of the given employees, three employees (Aniket and Ekant for November and Gorak for December) satisfy the given criteria. Ans: (3)

Solutions for questions 5 to 10:

From the flow chart, we can classify which requests will be approved and which will be rejected.

Let us consider employees from Delhi. If the employee has only an MBA (without BE), and he is not from Finance, the request will be approved. Hence, there will be zero rejected requests which satisfy the above conditions. Similarly, if the employee is from Delhi, has only MBA and is from Finance, the request will be denied. Hence, there will be zero approved requests which satisfy the above conditions.
Similarly, we can map which requests will be approved and which will be denied. We can get the below table for different combinations. In the table, all the cells with zero means that there cannot be any requests from that category.

		MBA			BE			Both		
		Fin	IT	Op	Fin	IT	Op	Fin	IT	Op
Delhi	Approved	0			0		0			
	Denied		0	0		0		0	0	0
Mumbai	Approved		0				0			0
	Denied	0		0	0	0		0	0	
Others	Approved	0	0		0	0		0	0	
	Denied			0			0			0

5. If a request of an employee with MBA in Operations was denied, then he cannot be an employee with only an MBA. Since all the requests from employees with only MBA working in Operations will be approved (from the above table). Hence, the employee must have both MBA and BE.
Also, if he is from Delhi or Other cities, his request will be approved (from the last column of the above table). His request will be denied only if he is from Mumbai.
Hence, the employee must be from Mumbai.
Choice (C)

6. Given that the request from an employee from Mumbai was denied. In the table above, this corresponds to the fourth row.
Option A: If the employee is from IT department, he cannot have a BE (since the cells corresponding to BE and both for IT Department are zeroes). This option is incorrect.
Option B: If the employee has an MBA, he can be from IT (in case of only MBA) or Operations (in case of both MBA and BE). Hence, this option is incorrect.
Option C: If he is a BE, he must be from Operations and not IT. Hence, this option is incorrect.
Option D: If he is from Operations department, he must have a BE (either only BE or both MBA and BE). Hence, this option is true. Choice (D)

7. The employee does not have a BE implies that he has an MBA.
Option A: He can work in the Finance department if he is from Mumbai. Hence, this statement need not be false.
Option B: He can work in Operations department, if he is from either Mumbai or Delhi. Hence, this statement need not be false.
Option C: If he works in the IT department, he must be from Delhi. Hence, this statement must be false and is the correct answer.
Option D: If he is from Delhi, he can be from IT or Operations. Hence, this statement need not be false.
Hence, option C is the answer. Choice (C)

8. From the given information, there are 180 requests from employees of each of the three departments. For each department, there were 60 requests from employees from the three cities (Delhi, Mumbai and Others). For each city, there were 20 from employees with only MBA, 20 from employees with only BE and 20 from employees with both.
For Delhi, 20 requests from employees with only MBA from Finance must all be rejected (since none can be accepted, from the above table). Similarly, 20 requests from employees with MBA from IT must all be accepted. In this manner, we can fill the above table, which is provided below.

		MBA			BE			Both		
		Fin	IT	Op	Fin	IT	Op	Fin	IT	Op
Delhi	Approved	0	20	20	0	20	0	20	20	20
	Denied	20	0	0	20	0	20	0	0	0
Mumbai	Approved	20	0	20	20	20	0	20	20	0
	Denied	0	20	0	0	0	20	0	0	20
Others	Approved	0	0	20	0	0	20	0	0	20
	Denied	20	20	0	20	20	0	20	20	0

The total number of requests denied during the given month = 60 + 60 + 120 = 240. Ans: (240)

9. Among the requests from employees with only an MBA, a total of 100 requests were accepted.
Among the requests from employees with both MBA and BE, a total of 120 requests were accepted.
Total number of requests accepted = 100 + 120 = 220. Ans: (220)

10. The number of requests rejected from employees of Finance = 100.
The number of requests rejected from employees of IT = 80
The number of requests rejected from employees of Operations = 60.
Hence, the maximum number of rejected requests were from employees of Finance. Choice (A)

Solutions for questions 11 to 14:

Let pillars from left to right be indicated by Pillar 1 through Pillar 7. From (v), Pillar 7 must be 20 ft shorter than Pillar 1. From the heights given the only possibilities for the heights of Pillar 1 and Pillar 7 are (30, 10) or (50, 30). If Pillar 7 is 10 ft high, then from (iii), the pillar to the immediate right of Pillar 7 must be 85 ft high, which is not possible. Hence, Pillar 7 must be 30 ft high, and Pillar 1 must be 50 ft high.
From (iv), Pillar 2 is 30 ft higher than Pillar 6. This is possible only if the heights of Pillar 2 and Pillar 6 are (75, 45) or (80, 50). However, since Pillar 1 is 50 ft high, Pillar 6 cannot be 50 ft high.
Hence, Pillar 2 must be 75 ft and Pillar 6 must be 45 ft. Neither of these pillars are Red.

From (iii), the 10 ft pillar is to the immediate left of the 85 ft pillar. And since neither are painted Blue, one of these must be painted Red.
If the 10 ft pillar is painted Red, the maximum possible height of the pillars painted Red will be 10 + 80 + 50 (since the 75 ft pillar, i.e., Pillar 2, is not Red) = 140.
Since it is given that the sum of the heights of the Red pillars is 10 ft more than the sum of the heights of the first three pillars, the sum of the heights of the first three pillars must not exceed 130 ft.
The first two pillars are 50 and 75 ft high. The minimum possible height of the third pillar can be 10 ft.
The minimum possible height of the first three pillars is 50 + 75 + 10 = 135.
Hence, if the 10 ft pillar is Red, condition (ii) can never be satisfied.
Hence, the 85 ft pillar must be Red.
The 85 ft pillar can be Pillar 4 or Pillar 5. However, from (v), Pillar 5 is Red. Since both Pillar 4 and Pillar 5 cannot be Red, the 85 ft pillar must be Pillar 5. Pillar 4 must be 10 ft high and must be Green.
Pillar 3 must be 80 ft high. The sum of the heights of the first three pillars is 50 + 75 + 80 = 205 ft.
The sum of the heights of the Red pillars must be 215 ft. We know that Pillar 5, which is Red, has a height of 85 ft. Hence, the other two pillars must have a combined height of 130 ft. This is possible only if the 50 ft and the 80 ft pillars are both Red.
Hence, Pillar 1 and Pillar 3 must both be Red. Since there must be at least two pillars of each colour, Pillar 7 must be Green (since it cannot be Blue from (v)).
Pillar 6 which is next to Pillar 7 cannot be Green and must be Blue. Since there are three red pillars, two green pillars, Pillar 2 must be Blue.

The following table provides the heights, colours, and order of the seven pillars:

Order	1	2	3	4	5	6	7
Height	50	75	80	10	85	45	30
Colour	Red	Blue	Red	Green	Red	Blue	Green

11. The height of the third pillar from the left is 80 ft. Choice (C)

12. The total height of all the pillars that are Blue = 75 + 45 = 120 ft. Choice (A)

13. The colour of the second pillar from the left is Blue. Choice (A)

14. The maximum difference in heights of any pair of pillar of the same colour is 35 ft (for the two red pillars with heights 85 ft and 50 ft). Choice (B)

Solutions for questions 15 to 20:

Given that each student can have one or two marbles with him initially. From this, we can find the number of marbles initially with the students and the number of marbles given. This is provided in the following table:

Student	Total Marbles	Initial Marbles	Marbles Given
A	2	1	1
B	4	1/2	3/2
C	5	1/2	4/3
D	2	1	1
E	5	1/2	4/3
F	4	1/2	3/2
G	6	2	4
H	3	1/2	2/1

From the given information, we can see that the last student would have received 4 green marbles. Among the given students, only three students, C, E and G, could have received 4 green marbles.

From (iii), G was standing in front of D. Hence, G could not have been the last person in the queue. Either E or C must have been the last person. And this student would have had one marble initially and received 4 green marbles.

From (ii), B was in the third place and B had 2 marbles initially and Kiran gave 2 green marbles. From (iii), the marbles that B initially had were Red.

Since B had two red marbles and Kiran gave two green marbles to B, the student behind B must have had two blue marbles.

The only students (apart from B) who could have two marbles initially are C, E, F and H. From (ii), H was not standing immediately behind B. From (iv), E was not in the fourth position.

Hence, only C or F can be behind B.

If C was behind B, and since C had 5 marbles, then from (i), the student in the seventh position must have had 2 marbles. This student must have initially had 1 marble and must have been given 1 marble. We also know that the person in the last place initially had 1 marble. Since these two persons (in the seventh and eighth places) had the same number of marbles, the person in the seventh place must have received two or three green marbles (based on the colours of marbles initially with them). Since this has a contradiction, this case is not possible.

Hence, F must be behind B. F must have had 2 blue marbles and must have been given 2 green marbles.

Since F, in the fourth place, had 4 marbles, the student in the seventh place must have had 3 marbles, in total. Among the students, H is the only one who has 3 marbles. Hence, H must have been in the seventh place.

From (iv), E was not adjacent to H. Hence, C must have been in the last place, and must have received 4 green marbles.

Also, since F has 2 green marbles, the student immediately behind F must have had 2 red marbles. Among the remaining students, the only student who could have had two marbles initially are E and G. If E is behind F, E must have had 2 red marbles initially and must have received 3 green marbles. Hence, the student behind E must also have had 2 red marbles initially. This can only be G. However, from (iii), G was standing in front of D. In this case, he will be in front of H. Hence, this is not possible.

G must have been behind F and must have had 2 red marbles. From (iii), D must have been behind G. Since G had 2 red marbles and received 4 green marbles, D must have had 1 blue marble and received 1 green marble. Since D received 1 green marble, H must have had 2 blue marbles and received 1 green marble. Since H received 1 green marble, C must have had 1 blue marble.

E could initially have had 1 or 2 marbles, while A had one marble.

If E is in the first place, and A is in the second place, then since A had one marble and received one marble, the colour of A's marble must have been the same as B's. Hence, A must have had 1 Red marble. If E is in front of A, and E received 3 green marbles, he must have had the same number of marbles as A and of the same colour. Since E had two marbles initially, this is not possible. If E received 4 green marbles, then E must not have the same number of marbles as A, in which case he must have 2 marbles initially. This is also not possible as E had a total of only 5 marbles. Hence, this case is not possible and A is in the first place, while E is in the second place.

A had one marble initially and received one green marble. Hence, the student behind A, i.e., E must have two marbles of a different colour. If E had two marbles, then he must have received three green marbles, which means he must have had the same number of marbles as B in the same colour. Hence, E must have had 2 red marbles and A must have had 1 red marble.

The following table provides the number of marbles of different colours for the eight persons and the order in which they are standing:

Order	8	7	6	5	4	3	2	1
Person	C	H	D	G	F	B	E	A
Total Marbles	5	3	2	6	4	4	5	2
Distribution	1B, 4G	2B, 1G	1B, 1G	2R, 4G	2B, 2G	2R, 2G	2R, 3G	1R, 1G

15. Kiran gave a total of 18 marbles to the students.
Choice (B)

$= -8+(538-538)+\ldots\ldots(538-538)+538 = 530$

∴ The remainder is 530

16. E had two red marbles with him initially. Choice (A)

17. D is standing immediately in front of H. Choice (C)

18. C had 1 blue marble and four green marbles. Choice (C)

19. The total number of marbles with the first four persons in the queue before Kiran gave them the marbles = 7. Ans: (7)

20. Total number of red marbles = 7
Total number of blue marbles = 6
Required difference = 1 Ans: (1)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	11, 12, 13, 14
Medium	1, 3, 4
Difficult	2, 5, 6, 7, 8, 9, 10, 15, 16, 17, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. When 1000 is divided by 1001, it leaves a remainder of –1
Now, 853853 .... (100 digits) = 8,538, ..., 538
Can be written as

$$8\left(10^{99}\right)+538\left(10^{96}\right)+538\left(10^{93}\right)+538\left(10^{90}\right)$$

$$+538\left(10^{3}\right)+538$$

$= 8(1000)^{33} + 538(1000)^{32} + ... + 538(1000)^{2} + 538(1000) + 538$
When this is divided by 1001, we get a remainder of

$$8(-1)^{33}+538(-1)^{32}+538(-1)^{31}+538(-1)^{30}+.....+538$$

Alternative Solution:

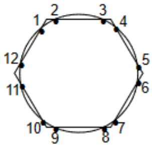
Since any six-digit number of the form *xyzxyz* is divisible by 1001 (because *xyz* × 1001 = *xyzxyz*), any number (whose number of digits is a multiple of six) of the form *xyzxyzxyz*...... upto, say, 6*k* digits will be divisible by 1001. Hence, 853853 .... upto 96 digits will be divisible by 1001.
Now, the required remainder is simply that of the last four digits, i.e., 8538, divided by 1001, i.e., 530. Ans: (530)

2. The total volume of water that flows through the pipes P_1 and P_2 per second $= \left\{2\left(\frac{15}{10000}\right)+6\left(\frac{25}{10000}\right)\right\} m^3 = 0.018\ m^3$.
Therefore, every second 0.018 m^3 of water will flow through the pipes into the tank.
Thus in 40 minutes, 40 × 60 × (0.018) m^3 or 43.2 kilolitres of water will be filled.
So the capacity of the tank is 43.2 kilolitres (since 1 m^3 = 1 kilolitre). Ans: (43.2)

3. Since they are running in the same direction with their speeds in the ratio *n* : 1, (and *n* is a natural number) they meet at (*n* – 1) distinct points. (At $\frac{L}{n-1}, \frac{2L}{n-1}, \frac{3L}{n-1}$ ....L, where L is the length of the track.) They meet at each of these points once every (*n* – 1) times they meet.
So, (*n* – 1) has to be a factor of the difference (17 – 5) or 12. Except for *n* = 6, for all other values of *n*, this is true. Choice (B)

4. p and q are the roots of f(x) = 0
Let $f(x) = ax^2 + bx + c$
$f(5) = 25a + 5b + c = 35 \rightarrow (1)$
$f(-5) = 25a - 5b + c = 15 \rightarrow (2)$
$(1) - (2) \Rightarrow 10b = 20 \Rightarrow b = 2.$
p + q = sum of roots = $\frac{-b}{a}$. However, as 'a' is still unknown $\frac{-b}{a}$ cannot be determined. Choice (D)

5.



As shown above, the numbers of points of intersection can be at most 12. [When the radius of the circle is greater than $\frac{\sqrt{3}}{2}$ times the side of the hexagon but less the side of the hexagon]

Ans: (12)

6. Let the instalment paid by Manishankar per year be denoted by *x*

Borrowed amount	Amount at the end of 1 year	Instalment paid	Balance
5460	$\frac{6}{5}(5460)$	x	$\frac{6}{5}(5460) - x$

Amount owned at the end of 1 year	Amount at the end of 2nd year	Instalment paid	Balance
$\frac{6}{5}(5460) - x$	$\frac{6}{5}\left(\frac{6}{5}(5460) - x\right)$	x	$\frac{6}{5}\left(\frac{6}{5}\times(5460) - x\right) - x$

Amount at the end of 2 year	Amount at the end of the 3rd year	Instalment paid	Balance
$\frac{6}{5}\left(\frac{6}{5}\times(5460) - x\right) - x$	$\frac{6}{5}\left(\frac{6}{5}\left(\frac{6}{5}\times(5460) - x\right) - x\right)$	x	0

$\therefore\ \frac{6}{5}\left(\frac{6}{5}\left(\frac{6}{5}\times(5460) - x\right) - x\right) - x = 0 \Rightarrow x = 2592$

Therefore, the instalment paid at the end of each year is ₹2592

Alternative Solution:

Using the present value of instalment approach we get $\frac{x}{1.2} + \frac{x}{(1.2)^2} + \frac{x}{(1.2)^3} = 5460.$

Now, solving the above equation using the online calculator, we get $x = 2592$.

Ans: (2592)

7. Since x is non-negative,
f(x) = 1/(1 + x)

$f(3) = \frac{1}{1+3} = \frac{1}{4}$

$\Rightarrow f^2(3) = (f(3)) = f\left(\frac{1}{4}\right) = \frac{1}{1+\frac{1}{4}} = \frac{4}{5}$

Similarly, we can show that

$f^3(3) = \frac{5}{9}$, $f^4(3) = \frac{9}{14}$ and $f^5(3) = \frac{14}{23}$

Hence, $f(3)\, f^2(3)\, f^3(3)\, f^4(3)\, f^5(3)$

$= \left(\frac{1}{4}\right)\left(\frac{4}{5}\right)\left(\frac{5}{9}\right)\left(\frac{9}{14}\right)\left(\frac{14}{23}\right) = \frac{1}{23}$

Choice (C)

8. Since –k is negative
f(x) = 1 + x
When *k* = 2,

$f(-2) = -1$
$f^2(-2) = f(-1) = 0$
and $f^3(-2) = f(0) = 1$
So, $f(-2) + f^2(-2) + f^3(-2) = 0$
In a similar way, we can show that $f^{k-1}(-k) + f^k$
$9 - k) + f^{k+1}(-k) = 0$ for any integer $k > 2$

Choice (B)

9. 17, 21, 25, …….. 417 and
16, 21, 26, …….. 466
21 is the first common term.
The terms in the first A.P. have a common difference of 4 and the terms in the second A.P. have a common difference of 5.
So, the common terms will also be in A.P. with a common difference of 4 × 5 = 20. We need to check for terms till 417.
So, $21 + 20(n - 1) \leq 417$, i.e., $21 + 20(19) = 401$ is the last common term and hence there are a total of 20 such common terms.

Sum of all the common terms $= 20 \times \left(\frac{21+401}{2}\right)$

$= 4220.$ Ans: (4220)

10. Let number of students in the batch be = x
Then number of male students are = 0.3x and the number of female students = 0.7x
10% of females have work experience
$\Rightarrow$ 0.07x females have work experience.
Using I alone, 0.2x have work experience.
$\Rightarrow$ The number of males who have work experience is 0.13x
Hence, I alone is sufficient.

Using II alone, the number of males having work experience is 1.25 times 0.07x = 0.0875 x.
Hence, II alone is also sufficient. Choice (A)

11. If $4x + 5y + 2z = 111$, then y must be odd.
Let $y = 2k - 1$, where k is a positive integer.
$\therefore 4x + 5(2k - 1) + 2z = 111$
$\Rightarrow 4x + 10k + 2z = 116$
$\Rightarrow 2x + 5k + z = 58$ ------- (1)
For every pair of (x, k) such that $2x + 5k < 58$, there will be a corresponding value of z, such that the equation (1) is satisfied.
Thus, we need to simply count the number of pairs of (x, k).
Now, for a fixed value of k, x can be at most $\left[\frac{57-5k}{2}\right]$, where $[x]$ is the greatest integer less than or equal to x.
Now, k can take values from 1 to 11.
When $k = 1$, we have x taking a maximum value of $\left\lfloor\frac{57-5}{2}\right\rfloor = 26$. Therefore, x can take values from 1 to 26, which gives a total of 26 solutions. These can be tabulated as follows:

k	1	1	1	 1
x	1	2	3	 26
z	51	49	47	 1

Proceeding in this manner, the number of solutions is given by $\sum_{k=1}^{11}\left[\frac{57-5k}{2}\right] = 26 + 23 + 21 + 18 + \ldots + 6 + 3 + 1$
Now, the sum of the series
$= (26 + 23) + (21 + 18) + \ldots + (6 + 3) + 1$
$= 49 + 39 + 29 + 19 + 9 + 1$
$= \frac{5}{2}[49 + 9] + 1 = 146$

Alternative solution:

$4x + 5y + 2z = 111$
As 111 is odd, $4x + 5y + 2z$ must also be odd, which implies y is odd.
Let $y = 2k - 1$, which gives $4x + 5(2k - 1) + 2z = 111$
$\Rightarrow 4x + 10k + 2z = 116$
$\Rightarrow 2x + 5k + z = 58$
Now, for $k = 1, 2, \ldots 11$, we get the equations $2x + z = 53$, $2x + z = 48$, $2x + z = 43$, and so on, respectively, till $2x + z = 3$.
In $2x + z = 53$, x can take values from 1 to 26, giving us 26 solutions.
In $2x + z = 48$, x can take values from 1 to 23, giving us 23 solutions.
In $2x + z = 43$, x can take values from 1 to 21, giving us 21 solutions.
Proceeding similarly, for $2x + z = 8$, x can take values from 1 to 3, giving us 3 solutions.
For $2x + z = 3$, x can take only one value, i.e., 1, giving us only one solution.
Therefore, the total number of solutions
$= 26 + 23 + 21 + 18 + \ldots + 6 + 3 + 1$
$= 49 + 39 + 29 + 19 + 9 + 1$
$= 146$
Therefore, $4x + 5y + 2z = 111$ has 146 solutions, such that x, y and z are positive integers.
Ans: (146)

12. Given xyz = 1

$\Rightarrow xy = \frac{1}{z}, \frac{1}{xy} = z$ ------------ (1)

$$\frac{1}{1+x+y^{-1}} + \frac{1}{1+y+z^{-1}} + \frac{1}{1+z+x^{-1}}$$

$$= \frac{y}{y+xy+1} + \frac{1}{1+y+xy} + \frac{1}{1+\frac{1}{xy}+\frac{1}{x}}$$

[$\because$ from equation (1)], from (1), We get

$$= \frac{y}{y+xy+1} + \frac{1}{1+y+xy} + \frac{xy}{xy+1+y}$$

$$= \frac{1+y+xy}{1+y+xy} = 1.$$ Choice (B)

13. $f(p, q, r)$ is equal to the minimum of $(p + r)$ and $(q + r)$ and will be $(p + r)$.
$g(p, q, r)$ is equal to either $(q - p)$ or $(p - r)$.
$f(p, q, r)$ may or may not be greater than $g(p, q, r)$.
Hence, choice (A) is not necessarily true.

$h(p, q, r) = \frac{f(p, q, r) + g(p, q, r)}{2}$
Choice (B) states that
$\frac{f(p, q, r) + g(p, q, r)}{2} < f(p, q, r)$
$\Rightarrow g(p, q, r) < f(p, q, r)$
Which may or may not be true.

Choice (C) states that
$= \frac{f(p, q, r) + g(p, q, r)}{2} < g(p, q, r)$
$\Rightarrow f(p, q, r) < g(p, q, r)$ which may or may not be true.

Alternative Solution:

By taking p = 1, q = 3, r = 1, option (B) gets eliminated.
By taking p = 2, q = 100, r = 1, options (A) and (C) also are eliminated. Hence, option (D).
Choice (D)

14. It is given that
$f(x) = f(f(x + 4))$, if $x < 100$ and $f(x) = x - 2$, if $x \geq 100$.
We start by taking values of x closest to 100, so that we get iterations of the function.

$f(99) = f(f(103)) = f(101) = 99$
$f(98) = f(f(102)) = f(100) = 98$
$f(97) = f(f(101)) = f(99) = 99$
$f(96) = f(f(100)) = f(98) = 98$

Proceeding similarly, we observe that for $x < 100$
$f(\text{odd}) = 99$ and $f(\text{even}) = 98$
$\therefore f(50) = 98$. Choice (A)

15. On solving the first two of the given equations we get $(x, y) = (1, 2)$. This point satisfies the third equation. Hence, the three lines pass through a common point. Therefore, the three lines are concurrent. Choice (C)

16. Given in MS
$2N = D(k) + 10$ ———— (1)
(where k is a natural number)
$\Rightarrow D > 10$ ———— (a)

Consider statement I:
$N = 2D(p) + 26$ ———— (2)
(where p is a natural number)
$\Rightarrow 2D > 26$, i.e., $D > 13$ ———— (b)
Now (1) – 2 × (2)
$\Rightarrow D(k - 4p) = 42$
$\Rightarrow D \times$ (some natural number) = 42
$\Rightarrow D = 42$ or a factor of 42 —— (c)
Now, considering the conclusions (a), (b) and (c) together, there is a set of possible values of D, where D = 42 or 21 or 14.
$\therefore$ There is no contradiction.
$\therefore$ MS is not inconsistent with I.

Consider statement II:
$N = D(m) + 17$ ———— (3)
(where m is a natural number)
$\Rightarrow D > 17$ ———— (d)
Now, (1) — 2 × (3)
$\Rightarrow D(k - 2m) = 24 \Rightarrow D \times$ (some natural number) = 24
$\Rightarrow D = 24$ or a factor of 24 —— (e)
Now, considering conclusions (a), (d) and (e) together, there is one possible value for D, where D = 24.
$\therefore$ There is no contradiction.
$\therefore$ MS is not inconsistent with II.

Hence MS is inconsistent with neither I nor II.

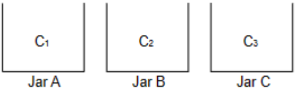
Since it is given that finally the concentration of exactly two of the three beakers is the same, either (i) the final concentration of B is equal to that of C, OR (ii) the final concentration of C equals that of A (Note that the concentration of A remains unchanged) OR (iii) the final concentration of A is equal to that of B. Now, since the initial concentrations of A and B are distinct, the final concentration of B cannot be equal to that of A.

Final concentration in Jar B = $\frac{C_1 + 2C_2}{3}$, which will lie between C_1 and C_2 and hence definitely not be equal to C_3.
Also, since in the last transfer, some of the contents of B are poured into C, whose concentration is C_3, (both contents of different concentrations), the final concentrations of B and C cannot be the same.
Hence, the only possibility left is that the new concentration of C = $\frac{\left(\frac{C_1+2C_2}{3}\right)+2C_3}{3}$ must be equal to the concentration of A, i.e., C_1.
Therefore $C_1 = \frac{C_1 + 2C_2 + 6C_3}{9}$
$\Rightarrow 8C_1 = 2C_2 + 6C_3$
$\Rightarrow 4C_1 = C_2 + 3C_3$. Choice (C)

18. Let p and q be two natural numbers. Then $p^2 - q^2$ is the quantity relevant to the question.
$p^2 - q^2 = (p + q)(p - q)$.
If $p + q$ and $p - q$ are odd, $(p + q)(p - q)$ is also odd. Hence any odd number can be expressed as the difference of two perfect squares.
If $(p + q)$ and $(p - q)$ are even, then $(p + q) \times (p - q)$ is a multiple of 4. Hence all multiples of 4 can be expressed as difference of two perfect squares.
If N is divisible by 2 but not by 4, then one of $(p + q)$ and $(p - q)$ is always even and the other is odd, which results in fractional values of p and q.
So, the numbers which are divisible by 2 but not by 4 cannot be written as a difference of two perfect squares.
[Eg: $38 = 19 \times 2$. Take $p + q = 19$ and $p - q = 2$, we get $p = 10.5$ and $q = 8.5$]
The number of odd natural numbers less than 1000 is 500.
The number of multiples of 4 less than 1000
$= \frac{1000}{4} - 1 = 249$.
$\therefore$ The required number of numbers = 500 + 249 = 749.

Choice (D)

17. Consider the three beakers A, B, C and their concentrations C_1, C_2 and C_3 respectively, where $C_2 < C_1 < C_3$.



Note that remaining numbers less than 1000 are multiples of 2 not divisible by 4. Choice (B)

19. $17\log_{30}x - 3(\log_x 5 + \log_x 6) + 20(\log_x 10 + \log_x 3)$
$17\log_{30}x - 3\log_x 30 + 20\log_x 30$
$= 17\log_{30}x + 17\log_x 30$
$= 17(\log_{30}x + \log_x 30)$
Since $x > 1$, $\log_{30}x > 1$
$\Rightarrow \log_{30}x + \log_{30}x \geq 2$ (using AM-GM inequality)
$\therefore$ Minimum value of the given expression is $17 \times 2 = 34$. Ans: (34)

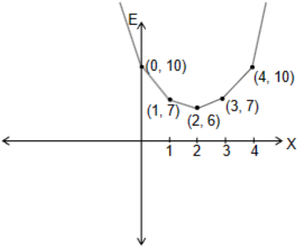
20. To find the integral values of x, we divide it in 3 ranges - $x \leq 0$, $x \in (0, 4)$ and $x \geq 4$.

Case (i)
When $x \geq 4$, all the values in modulus are positive.
$E = x + x - 1 + x - 2 + x - 3 + x - 4$
$= 5x - 10$
$5x - 10 < 54$
$x < 12.8$
$\therefore x = \{4, 5, 6, 7, 8, 9, 10, 11, 12\} = 9$ values.

Case (ii)
When $x \leq 0$, all values are negative (or) zero.
$E = -x + 1 - x + 2 - x + 3 - x + 4 - x$
$= 10 - 5x$
$10 - 5x < 54$
$x > \frac{-44}{5}$
$\Rightarrow x > -8.8$
$\therefore x = \{-8, -7, -6, -5, -4, -3, -2, -1, 0\} = 9$ values.

Case (iii)
$x \in (0, 4)$
The integer values of x in this range are x = 1, 2, 3.
For x = 1 & 3, E = 7 and when x = 2, E = 6.

The graph of E looks like



21. Let $\angle PRQ = x°$
$\angle PQR = x + 30°$
$\angle QPR + \angle STU = 220°$
In ΔPQR,
$\angle PQR + \angle QPR + \angle PRQ = 180°$
$\angle QPR = 180 - (x° + x° + 30°) = 150° - 2x$
$\angle STQ + \angle STU + \angle UTR = 180°$ ------ (1)
($\because$ QTR is a straight line)
$SQ = ST \Rightarrow \angle STQ = \angle SQT = x° + 30°$
$TU = UR \Rightarrow \angle UTR = \angle URT = x°$
$(1) \Rightarrow \angle STU = 180° - (x° + 30° + x°)$
$= 150° - 2x°$
$\angle QPR + \angle STU = 220°$
$2(150° - 2x°) = 220°$
$x° = 20°$ Choice (D)

22. If the smaller of two numbers is a and the greater is b, the possible values of a and the corresponding values of b (such that $a + b \geq 20$) and the number of ordered pairs (a, b) are given below.
If a = 1, 3, 5, 7, 9, there are respectively 1, 2, 3, 4, 5 values of b such that $a + b \geq 20$, i.e. a total of 15 pairs (a, b).
If $a \geq 11$, there are $^5C_2$ is i.e. 10 values of (a, b).
Thus, in all, there are 25 pairs (a, b) such that $a + b \geq 20$. Choice (A)

Difficulty level wise summary - Section III

Level of Difficulty	Questions
Very Easy	–
Easy	2, 4, 10, 12, 15
Medium	5, 9, 13, 14, 17, 18, 19, 21, 22
Difficult	1, 3, 6, 7, 8, 16, 20
Very Difficult	11

∴ There are 21 values of x which satisfy the condition. Ans: (21)

Done
Tip: Scroll slowly for the best results